Chapter 11 exercises

DS11-01 • B-tree insertion and deletion

Use minimum degree t=2 and top-down splitting. From empty, insert 10,20,5,6,12,30,25, then delete 12, preferring the left sibling when borrowing. Show each split, promotion, borrowed key, and final tree. Check every occupancy requirement.

DS11-02 • Height and I/O

Derive $n \geq 2t^h - 1$ for a nonempty B-tree with height h≥1. For t=64, n=1,000,000, find the greatest integer height permitted by this bound. Does that inequality exclude height 2? Use maximum capacity as a separate check.

DS11-03 • B+ separators and deletion

With maximum leaf capacity L=3, insert 10 into `[4,9,12]` and split into two records per leaf. Use right-minimum separators. Draw the root, leaf chain, and record locations. In a separate legal L=2 state with root `[10]` and leaves `[4,9]`, `[10,12]`, delete 10. Which keys and links change?

DS11-04 • Range-query assumptions

B+ leaves hold keys and record pointers. Integer capacities M≥3 and L≥2 bound internal child counts and leaf record counts. Their nonroot minima are a=ceil(M/2) and ell=ceil(L/2). An internal root has at least two children. Height h counts root-to-leaf edges. Derive general height and index I/Os for k outputs, then state conditions yielding $O(1 + \log_B n + k/B)$. Separate scattered/clustered full records and output work. Construct the M=3,$L = B = 2^q$ full-binary counterexample. Explain why the physical location of child.minimum and persistent parent fields affects update I/Os.

DS11-05 • Incompatible order conventions

Text A defines order m as maximum children. Text B defines minimum degree t as at least t and at most 2t nonroot children. With m=5, can t=m/2 be substituted into this chapter's split procedure? State capacity and rounding differences.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

Chapter 11 exercises (continued)

DS11-06 • B+ underflow, exact separators, and a range

Let M=3,L=3. Root separators are [10,14], with leaves [4,9]↔[10,12]↔[14,16]. Delete 10,14,4 in order. Show each borrow or merge, parent separators, minimum occupancy, and leaf links. Prefer merging with the left sibling at the first step. After deleting 14, execute inclusive range [9,14], distinguishing inspected keys from output. Explain final root contraction and whether an intervening B-tree parent key could also be discarded.

DS11-07 • Internal underflow, child transfer, and root contraction

> M=4,L=3; separators are exact right minima. Internal entries are (child_id,cached_min). Root R has P,Q. P owns leaves A=[1,3], B=[5,7]; Q owns C=[9,11], D=[13,15], E=[17,19]. The chain is A↔B↔C↔D↔E and edge-height is two. Delete 1. Try a same-parent sibling before merging; retain the left page on merge.

1. Trace the leaf merge and internal borrowing. Give P.entries,Q.entries,root entries, all separators, and the leaf chain. Trace a search for 9.
2. Independently start without E and delete 1. Give the internal merge and contracted root. What does closed range [7,14] output?
3. A proposed merge also inserts the old root separator as a data record into a leaf. Which key would be duplicated? Check the correct child and separator counts.

References and attribution

Ellis Horowitz, Sartaj Sahni, Dinesh Mehta. Fundamentals of Data Structures in C++, 2nd Edition.

Original educational synthesis by Lin Cing. Original material is licensed under CC BY 4.0. The referenced textbook is not distributed with these notes.

https://github.com/xinhi6716/computer-science-self-study